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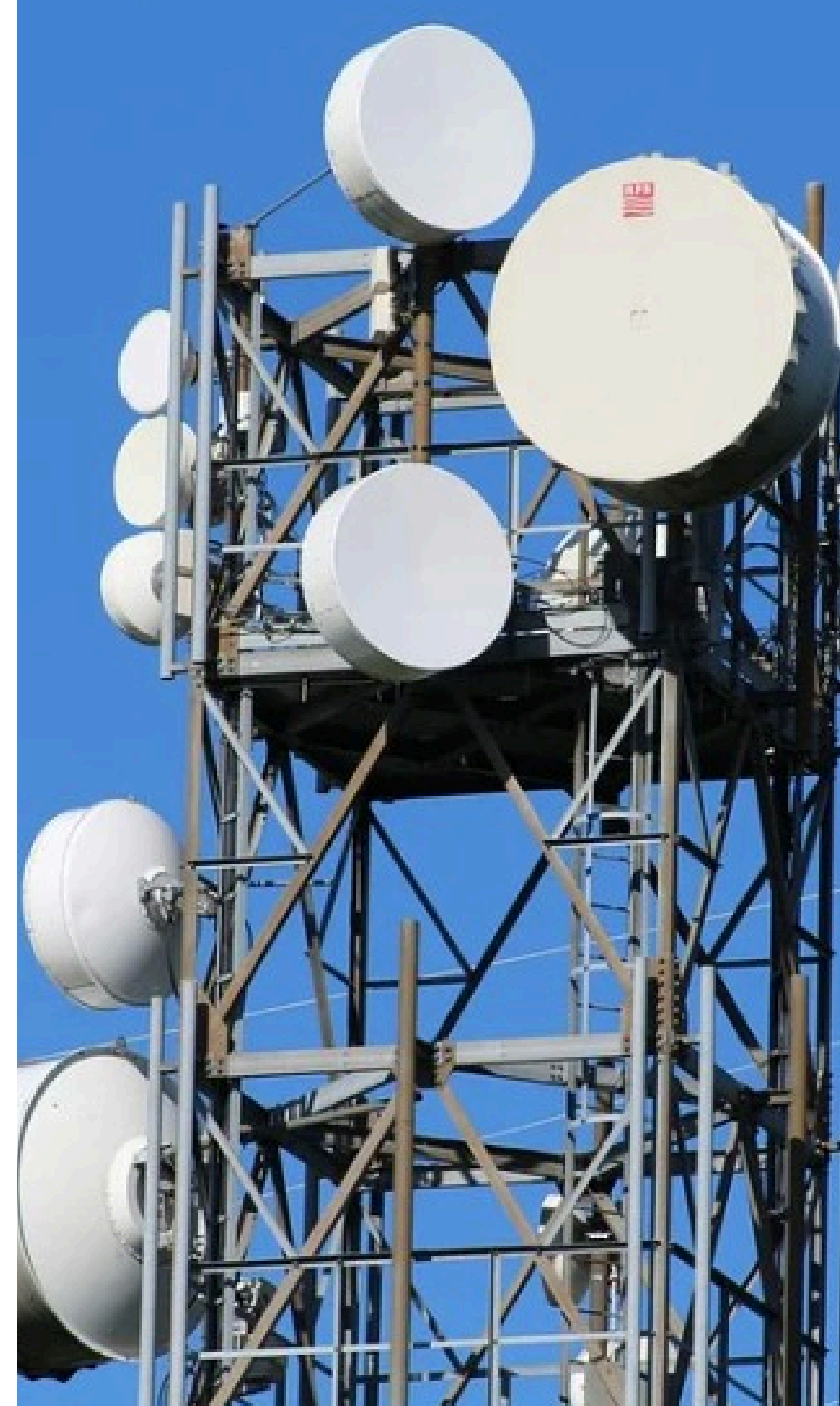
A Telecom Company Case Study

Reducing Customer Churn Through Predictive Analytics



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Market Context : Telecom Churn Costs the Industry Billions And Most Companies Still Can't Predict It

Company A is losing ~49,600 customers, this represents an estimated \$29M+ in annual recurring revenue at risk.¹

**(49,562 churned × \$58.72 avg monthly rev × 12 months)*

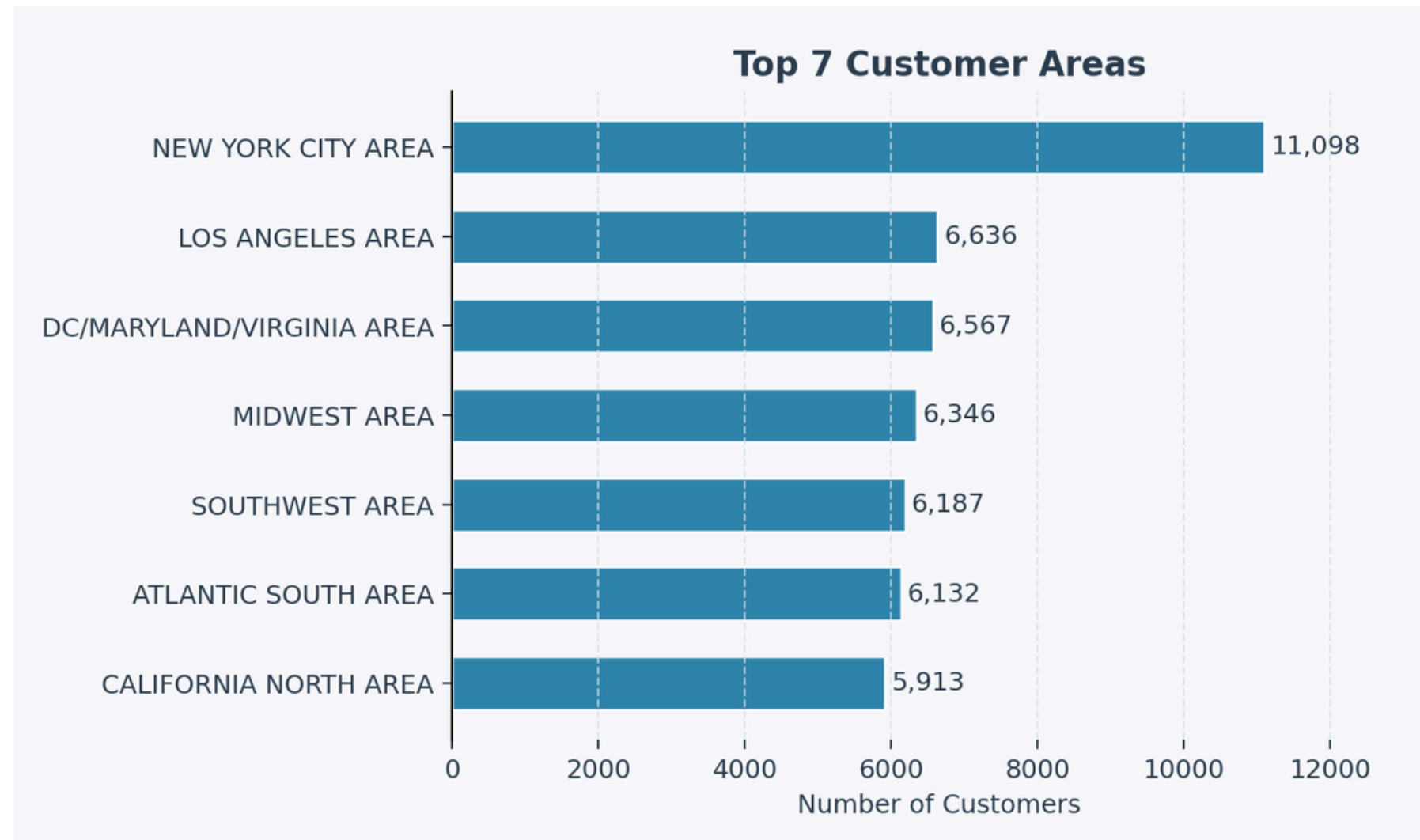
Competitors are increasingly offering better pricing and services, accelerating customer migration. Companies without a predictive system react too late, after the revenue is already lost.

¹ *Company A internal dataset, 2026 — authors' own analysis*

Client Situation: Company A Serves 100,000 U.S. Customers

Company A's customer base is concentrated in major metropolitan areas of the United States, with a predominantly solvent and home-owning profile, representing a high-value segment worth retaining.

Client Situation

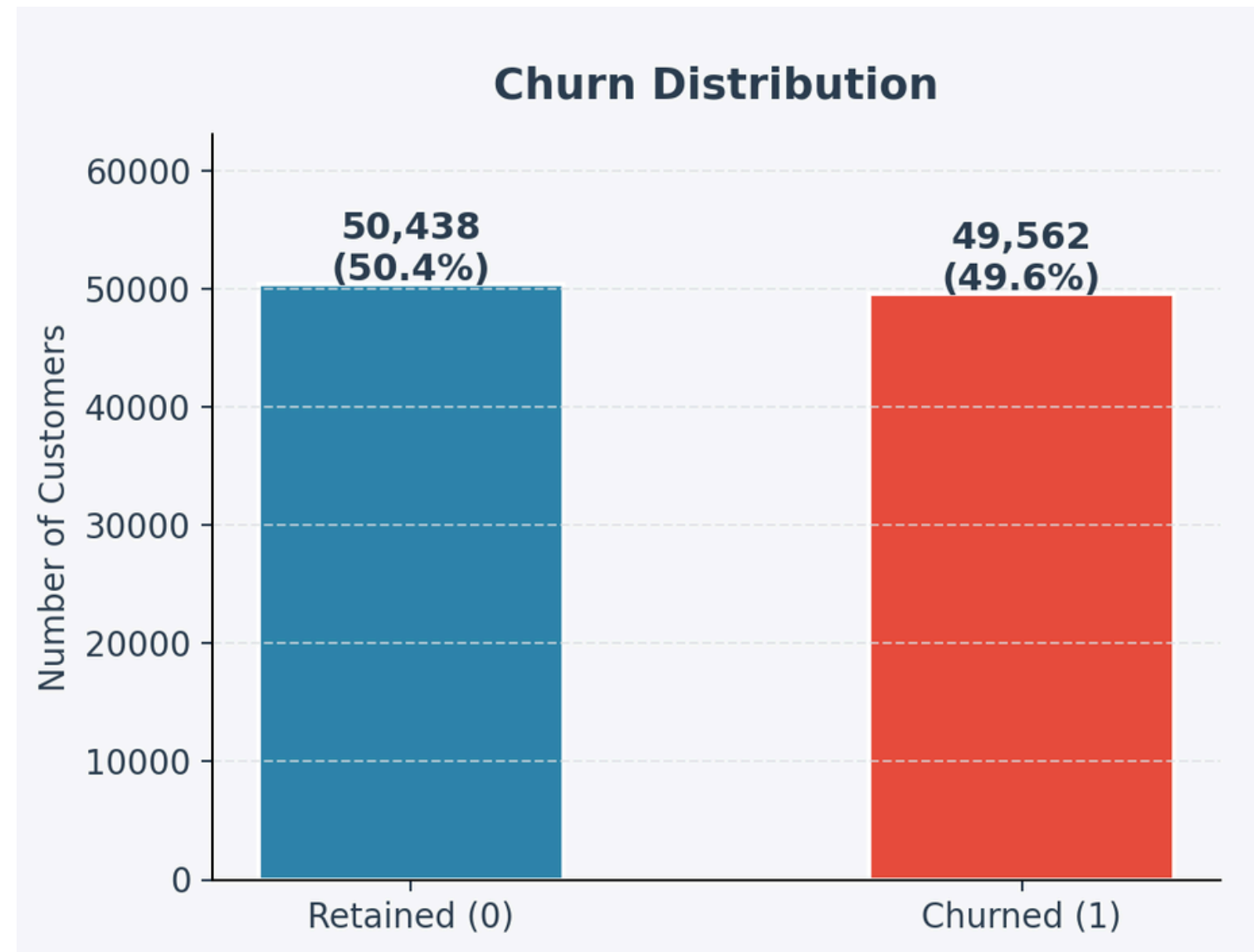


Customers are concentrated in the top U.S. accountants, with New York City leading the way at 11,098, nearly double that of any other market.

Customer Profile	
Avg Lifetime Revenue	\$1,031.92
Avg Equipment Age	392 days (~1.1 yrs)
Credit Card Holders	67%
Homeowners	97% of known
Top Credit Class (AA)	36.5%
New Cell Users	19.3%

A financially stable base: average lifetime income of \$1,031, 67% credit card holders and 36.5% in the highest credit class (AA).

Client Situation



Nearly 1 in 2 customers cancelled their subscriptions (49.6%), well above the industry average, which ranges between 15% and 25%.

Company A has a high-value customer base, but it is losing it at an alarming rate, with no predictive system in place to prevent it.

What the data shows - Part 1:

Who Is Likely to Churn?

Variable	Churner Profile	Interpretation
Equipment Age	Older devices	Upgrade frustration and outdated technology.
Tenure	Shorter tenure	Customer loyalty has not yet been established.
Single Device Accounts	More common	Lower switching costs than multi-device households.
Inactive Subscriptions	Slightly higher	Indicates lower engagement with services.
Device Condition	More refurbished devices	Indicates lower willingness to invest in premium devices.
Handset Price	Lower-cost devices	Lower device investment often reflects lower commitment to the provider.

According to Ribeiro et al. (2024)², long-term customers have a lower churn rate due to greater loyalty and depth of relationship with the operator, a dynamic that is observed in Company A's own data.

² Ribeiro, H., Barbosa, B., Moreira, A. C., & Rodrigues, R. G. (2024). Determinants of churn in telecommunication services: A systematic literature review. *Management Review Quarterly*, 74(3), 1327–1364. <https://doi.org/10.1007/s11301-023-00335-7>

What the data shows - Part 2:

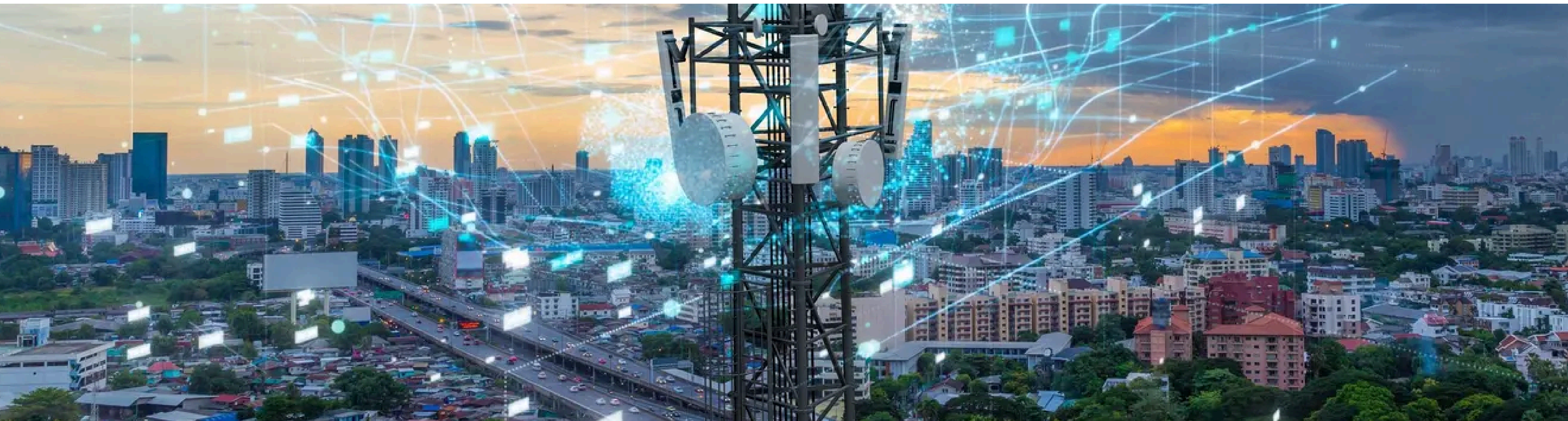
How They Behave Before Leaving

Variable	Churner Profile	Interpretation
Drop Rate	Higher drop rate	Poor call quality leads to dissatisfaction.
Monthly Revenue	Lower monthly spending	Spending declines as engagement weakens.
Average Usage	Lower average usage	Indicates decreasing dependence on the service.
Revenue per Minute	Lower revenue per minute	Indicates lower service value and usage.
Usage Trend	Negative trend	Consistent reduction in activity is a strong warning
Monthly Revenue	Lower monthly spending	Spending declines as engagement weakens.

According to Mismar et al. (2026)³, QoE indicators, such as high call drop rates, low connection rates, higher billing surcharges, and a progressive decline in both usage and revenue, would provide stronger churn signals than traditional network counters alone.

³ Mismar, F. B., Saleh, A., Pasaribu, I. M. P., & Syaifuddin, S. (2026). *From Network Experience to Subscriber Retention: An Explainable AI Framework for Mobile Operators*. arXiv.

Company A already has everything it needs to identify high-risk customers before they leave. What's needed is a model that connects the dots.



Our Approach

Using the company's customer data, a CatBoost-based churn prediction model was developed to identify customers at risk of leaving.

The model achieved a Recall of 70.3%, capturing more than seven out of every ten actual churners.

It also delivered an AUC of 0.653 and an F1-score of 0.634, indicating a solid balance between identifying churners and limiting false alarms.

Notes

Metric	Meaning
Accuracy	Overall prediction correctness.
Precision	Correctly identified churners among predicted churners.
Recall	Churners successfully detected by the model.
F1-Score	Balance between Precision and Recall.
ROC AUC	Ability to distinguish churners from non-churners.

Note: The proposal established the following performance targets for the churn prediction model: AUC (cross-validation) > 0.60, AUC (test) > 0.60, Recall > 0.60, and F1-score > 0.60.

Results

Metric	Target	Result
AUC (CV)	> 0.50	0.63
AUC (Test)	> 0.50	0.63
Recall	> 0.50	0.75
F1-Score	> 0.50	0.64
Precision	> 0.50	0.565

Notes

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Accuracy	Overall prediction correctness.
Precision	Correctly identified churners among predicted churners.
Recall	Churners successfully detected by the model.
F1-Score	Balance between Precision and Recall.
ROC AUC	Ability to distinguish churners from non-churners.

		PREDICTED LABEL	
		NO CHURN (0)	CHURN (1)
ACTUAL LABEL	NO CHURN (0) Actual Non-Churners	✓ 7,452 Correctly Identified Non-Churners (True Negatives)	✗ 7,679 False Alarms (False Positives)
	CHURN (1) Actual Churners	✗ 4,412 Missed Churners (False Negatives)	✓ 10,457 Correctly Identified Churners (True Positives)

The model prioritizes churn detection while maintaining acceptable prediction quality, making it suitable for supporting targeted customer retention initiatives.

What this Means for the Business

- Early identification of at-risk customers, enabling proactive retention actions before churn occurs.
- More efficient allocation of retention budgets by targeting customers with the highest likelihood of leaving.
- Reduced revenue loss through improved customer retention strategies.
- Improved customer lifetime value (CLV) by retaining profitable customers for longer periods. ⁴

⁴ Gattermann-Itschert, T., & Thonemann, U. W. (2022). Proactive customer retention management in a non-contractual B2B setting based on churn prediction with random forests. *Industrial Marketing Management*, 107, 134–147. <https://doi.org/10.1016/j.indmarman.2022.09.023>



The Proposal & Expected Impact

Implement a CatBoost-based customer churn prediction model within your customer management workflow to detect churn risk and to define retention strategies.

Expected Bussines Impact

The churn prediction model identifies 4,096 high-risk customers, representing approximately \$236,223 in monthly revenue at risk. By targeting only the top 20% of customers ranked by churn risk, the company can focus retention efforts on a segment that accounts for 27.5% of total revenue exposure, improving the efficiency and impact of retention campaigns.

Future Steps

- Explore ensemble approaches that combine multiple predictive models to improve churn detection performance.
- Incorporate new customer behavior and service quality variables to enhance model accuracy.
- Monitor model performance and retrain periodically to adapt to changing customer behavior patterns.
- Measure the business impact of retention campaigns driven by model predictions.

The next phase focuses on transforming predictive insights into measurable retention and revenue outcomes.

References

1. *Company A internal dataset, 2026*
2. *Ribeiro et al. (2024). Determinants of churn in telecommunication services: A systematic literature review. Management Review Quarterly, 74(3), 1327–1364. <https://doi.org/10.1007/s11301-023-00335-7>*
3. *Mismar, F. B. et al. (2026). From Network Experience to Subscriber Retention: An Explainable AI Framework for Mobile Operators. arXiv.*
4. *Gattermann-Itschert, T., & Thonemann, U. W. (2022). Proactive customer retention management in a non-contractual B2B setting based on churn prediction with random forests. Industrial Marketing Management, 107, 134–147. <https://doi.org/10.1016/j.indmarman.2022.09.023>*